Local Residual Smoothing and Exact Conformal Validity: a Small-Sample Separation

Peter Cotton
peter.cotton@microprediction.com

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Abstract

Fix any model, and consider the remaining task: estimate the conditional law of its residual at a test point from held-out residuals. When that law varies smoothly with the input, the efficient estimator at a small sample size borrows from neighbours by kernel smoothing, with a bandwidth that widens as data thin. We show that this borrowing is incompatible with an exact, distribution-free, finite-sample coverage guarantee. Any test-centred smoothing incurs a coverage gap of the same order as its bias, so exact coverage forces the bandwidth to its widest, uniform pooling, which is the crudest estimate of the residual law. The two goals, efficient small-sample estimation and an exact guarantee, are separated by a quantity of order h^s in the bandwidth h, and the estimation-optimal bandwidth makes that quantity largest exactly when data are scarce. The separation vanishes only as the sample grows or when the residual law does not vary with the input. Nothing here concerns the base model, which may be arbitrarily adaptive; the tension is entirely in the pooling step that supplies the guarantee.

1 The task after the model

Grant the strongest version of the standard defence of conformal prediction: the base model may be anything, a gradient-boosted tree, a normalizing flow, a neural quantile network, fit with full access to labels on a training fold. Freeze it. It induces a scalar residual, or nonconformity score, $R = A(X, Y)$, and the only remaining question is how to turn a held-out sample of residuals into a predictive statement at a fresh input. That step is where a conformal guarantee is earned or lost, and it is the only step we analyse. Everything below is downstream of an arbitrarily good model.

Write $\mu_x = \mathcal{L}(R \mid X = x)$ for the conditional residual law and let $(X_i, R_i)_{i=1}^n$ be an exchangeable calibration sample. We want a predictive law, or a prediction set, for the residual at a test point x_0, and we ask two things of it: that it estimate μ_{x_0} efficiently, and that its set have coverage $1 - \alpha$. We work on a metric input space $(\mathcal{X}, \rho)$ of dimension d under two assumptions that are mild and, for the interesting case, both present.

Assumption 1 (Smoothness)**.** The residual law is Hölder-s in total variation: there are $C < \infty$ and $s > 0$ with $d_{\mathrm{TV}}(\mu_x, \mu_{x'}) \le C\,\rho(x, x')^s$ for all x, x'.

Assumption 2 (Heterogeneity)**.** The residual law depends on the input: $\mu_x \ne \mu_{x'}$ for some x, x'.

Assumption 2 is what makes the problem nontrivial; without it a single pooled law is correct and there is nothing to estimate locally. Assumption 1 is what makes borrowing from neighbours pay: nearby inputs have nearby residual laws, so a neighbour's residual is informative about the test point.

2 The efficient estimator borrows from neighbours

The natural estimator of μ_{x_0} from the sample is a weighted empirical law

$$\widehat{\mu}_w(x_0) \;=\; \sum_{i=1}^{n} w_i(x_0)\,\delta_{R_i}, \qquad w_i(x_0) \geq 0, \quad \sum_i w_i(x_0) = 1, \tag{1}$$

and the natural weights are a kernel, $w_i(x_0) \propto K\big(\rho(x_0, X_i)/h\big)$, with a bandwidth h that sets how far the estimator reaches for neighbours. Under Assumptions 1–2 the bias of (1) in total variation is of order h^s, the variance of order $1/(nh^d)$, and balancing them gives the familiar minimax bandwidth and risk

$$h^* \asymp n^{-1/(2s+d)}, \qquad \mathbb{E}\, d_{\mathrm{TV}}\big(\widehat{\mu}_w(x_0), \mu_{x_0}\big) \asymp n^{-s/(2s+d)}, \tag{2}$$

which is the optimal rate for estimating a Hölder-s law in d dimensions (Tsybakov, 2009). The point we need from (2) is not the rate but its dependence on scale: at a small effective sample size m the optimal bandwidth is *large*, $h^*(m) \asymp m^{-1/(2s+d)}$, because with few points one must reach further to average anything at all. Efficient estimation of a smoothly varying residual law at small m requires wide, test-centred borrowing.

3 Exact coverage forbids the borrowing

Now ask the weighted estimator (1) to carry a coverage guarantee. The prediction set is the weighted $(1-\alpha)$ region of $\widehat{\mu}_w(x_0)$. Two facts bound what its coverage can be.

Lemma 1 (Localization opens a coverage gap)**.** *Let $C_\alpha(x_0)$ be the level-$(1-\alpha)$ set of the weighted law* (1)*. Then, writing $\widetilde{w}_i$ for the normalized weights including the test slot,*

$$\mathbb{P}\big(R_0 \in C_\alpha(x_0)\big) \;\geq\; 1 - \alpha \;-\; \sum_{i=1}^{n} \widetilde{w}_i(x_0)\, d_{\mathrm{TV}}\big(\mu_{X_i}, \mu_{x_0}\big), \tag{3}$$

and the bound is attained in the worst case over laws satisfying the stated total variations. Under Assumption 1 with kernel weights of bandwidth h,

$$\sum_i \widetilde{w}_i(x_0)\, d_{\mathrm{TV}}\big(\mu_{X_i}, \mu_{x_0}\big) \;\leq\; C \sum_i \widetilde{w}_i(x_0)\, \rho(x_0, X_i)^s \;=\; \Theta\big(h^s\big).$$

Proof. Inequality (3) is the coverage bound for non-exchangeable weighted conformal prediction (Barber et al., 2023): replacing the exchangeable reference by a covariate-weighted one costs, in coverage, the weighted total-variation distance between the calibration and test residual laws. The second display substitutes Assumption 1 and evaluates the kernel-weighted average of $\rho(x_0, X_i)^s$, which under a bandwidth h is of order h^s since the weights concentrate on $\{\rho(x_0, X_i) \lesssim h\}$. □

The kernel weights are informative, and so are non-uniform in a way that depends on the test point x_0; that is exactly what (3) charges for. The only way to drive the gap to zero, absent a known and fixed covariate-shift tilt of the kind studied by Tibshirani et al. (2019), is to remove the test-centred dependence of the weights, that is, to make them uniform within a region pre-committed before x_0 is seen. We record the endpoint.

Proposition 2 (Exact validity forces uniform pooling)**.** *Suppose the predictor* (1) *attains* $\mathbb{P}(R_0 \in C_\alpha(x_0)) \geq 1-\alpha$ *for every law satisfying Assumptions 1–2 with the stated constants, with no further assumption on the covariate distribution. Then the localization term in* (3) *is zero, which under Assumption 2 requires the weights to be constant across any pair of calibration points with* $\mu_{X_i} \neq \mu_{X_j}$*. Equivalently, the estimator is the equal-weight empirical law within a stratification fixed independently of* x_0*: uniform pooling, the* $h \to \infty$ *end of* (1)*.*

Proof. By Lemma 1 the worst-case coverage is $1-\alpha$ minus the localization term, so distribution-free validity at level $1-\alpha$ forces that term to vanish. It is a weighted average of nonnegative total variations $d_{\mathrm{TV}}(\mu_{X_i}, \mu_{x_0})$; under Assumption 2 some of these are strictly positive for calibration points in distinct residual regimes, so the term vanishes only if the weights place no differential mass across such points. A weight vector that is constant across all covariate-distinct points and independent of x_0 is uniform pooling within the pre-committed partition into residual-law level sets. $\square$

4 The separation

Put the two sides together. Efficiency wants the bandwidth at $h^*(m)$, wide when the effective count m is small; validity wants the bandwidth at infinity. They meet only at $h^* = \infty$, that is, only when there is nothing to gain from localization.

Theorem 3 (Small-sample separation)**.** *Under Assumptions 1–2, no predictor of the form* (1) *attains both*

(a) exact distribution-free coverage, $\mathbb{P}(R_0 \in C_\alpha(x_0)) \geq 1-\alpha$ *for every law with the stated constants; and*

(b) the minimax estimation rate (2) *for* μ_{x_0}*,*

unless the sample is unbounded or the residual law is input-free. Quantitatively, along the bandwidth family the coverage gap and the excess estimation risk trade at the same order: a predictor with estimation risk within a constant of the minimax rate $m^{-s/(2s+d)}$ *has coverage gap* $\Theta\big(h^*(m)^s\big) = \Theta\big(m^{-s/(2s+d)}\big)$*, while a predictor with zero coverage gap has estimation risk bounded below by the heterogeneity of the residual law across the coarsest admissible pool, a constant that does not vanish as* $m \to 0$*. The product of the two deficits is bounded away from zero for every finite* m*; it tends to zero only as* $m \to \infty$*.*

Proof. By Proposition 2, (a) forces uniform pooling, whose bias is $\sup\{d_{\mathrm{TV}}(\mu_x, \mu_{x'})\}$ over the pooled region, a positive constant under Assumption 2 independent of m; this violates (b), whose risk (2) tends to zero in m. Conversely, any predictor achieving (b) uses a bandwidth $h \asymp h^*(m)$, and by Lemma 1 its coverage gap is $\Theta(h^*(m)^s)$, positive for finite m, so it fails (a). The trade at a common bandwidth follows from (2) and Lemma 1: bias and coverage gap are both $\Theta(h^s)$, and reducing one raises the other. As $m \to \infty$, $h^*(m) \to 0$ and both deficits vanish. $\square$

Corollary 4 (The scarce-data regime)**.** *The separation is monotone decreasing in the effective sample size and maximal as $m \to 0$. In the regime where local smoothing most improves the estimate, small m under Assumption 1, the coverage gap of the estimation-optimal predictor is largest, and the estimation penalty of the exactly-valid predictor is its full heterogeneity constant.*

5 What the separation says

The base model is not on trial. It may borrow, adapt, and use every label it is given, because it is fit on a disjoint fold and the analysis begins after it is frozen. The tension is entirely in the pooling of held-out residuals that supplies the guarantee. There the choice is stark and quantified. To estimate a smoothly varying residual law well from a small sample one must weight nearby residuals more, and that weighting, by (3), costs coverage in proportion to how far the estimator reaches and how much the residual law changes over that reach. To keep the guarantee exact one must weight all residuals in a pre-committed region equally, which is the widest possible reach and the largest possible bias. The efficient thing and the guaranteed thing are the two ends of one bandwidth, and at a small effective sample size they are far apart.

This is why, with effective sample sizes well below a thousand, a kernel or local-likelihood estimate of the residual law is usually the better forecast and does not come with, and is not improved by, an exact conformal wrapper. The wrapper would either localize, and forfeit the exactness that is its entire justification, or pool, and forfeit the estimate. The separation closes only as data accumulate, when a narrow bandwidth is both efficient and nearly exchangeable, or when the residual law does not vary with the input, when there was never a reason to localize. Between those limits, in the small-sample heterogeneous regime that describes most applied forecasting, the guarantee and the estimate pull in opposite directions, and the size of the disagreement is h^s.

References

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